Credit Assignment or Observability?

Separating two causes of coordination failure in decentralized multi-robot task allocation under bursty communication

ROB 538 - Multiagent Systems | Prof. Kagan Tumer
Atharva Jadhav

The question

When a multiagent system underperforms because its agents act on local information, two very different things can be wrong. An agent may be unable to tell **what it caused** - its contribution buried in a shared reward, the credit assignment problem that difference rewards are built to fix. Or it may be unable to tell **what is happening** - the information needed to act well simply absent from its observations, which no amount of reward shaping can recover. The two are easy to conflate because they produce the same symptom. I propose to study a domain where they can be varied independently and measured against each other.

Domain

Robots are assigned tasks and some fail mid-execution; the team must detect the failure and reallocate the task over a channel with correlated, bursty packet loss. Each agent sees only the heartbeats it has received, and decides independently whether to take over a teammate's task. System utility is tasks completed, less duplicated effort, less detection latency. The environment will be an event-driven simulator with a Gilbert-Elliott burst channel and teams of 6 to 32 agents, run for 10^2-10^3 trials per configuration.

The coordination failure to be explained

Silence from a teammate is ambiguous: it may mean the teammate has died, or that the channel is in an outage. Taking over a dead teammate's task raises system utility; taking over a live one lowers it, by putting two robots on one task. During a burst, every agent independently concludes that every teammate has died, and the team produces a storm of duplicate takeovers that grows with team size - a scaling problem of the kind this course treats. The question I want to answer is which of the two causes above produces it.

Experiment: a 2x2 that separates the causes

I will train agents to learn a takeover policy from local observations, varying two factors independently:

> **Reward signal** - global team utility G, versus a difference reward D_i = G(z) - G(z_-i + c_i), the counterfactual "what would system utility have been had I not taken over." D_i is exactly computable in this domain, because a run can be replayed with one agent's action replaced by a default.
>
> **Observation** - the agent's own received heartbeats only, versus those plus a measure of how much of the *rest* of the team it can currently hear.

If the difference reward alone removes the storm, the problem was credit assignment. If the added observation alone removes it, reward shaping cannot substitute for a missing signal. If both are needed, the effects are separable, and the 2x2 says by how much.

Hypothesis

I expect observability to dominate, for a structural reason: during an outage the agent's observations genuinely do not distinguish a dead teammate from an unheard one, so the information required to choose correctly is absent rather than merely obscured. A difference reward can sharpen an agent's estimate of its own contribution, but it cannot add a distinction the observation does not carry. If that holds, the result is a clean negative for the difference-reward toolkit wherever the limiting factor is what an agent can see rather than what it can be credited for - worth diagnosing before choosing a method. If credit assignment instead carries most of the effect, that is the more surprising outcome and the more interesting one.

Evaluation and baselines

I will measure false takeovers, detection latency and wasted travel against team size from 6 to 32 agents, with learning curves for each cell of the 2x2. Baselines: a hand-designed rule gating its failure belief on team audibility, the phi-accrual detector from distributed systems, and no recovery. I also intend to implement a single-item auction among audible agents, testing whether explicit negotiation beats learned implicit coordination when the channel is the bottleneck.